

```
#include <iostream>
```

```
#include <string>
```

```
#include <vector>
```

```
using namespace std;
```

```
class Person {
```

```
protected:
```

```
    int id;
```

```
    string name;
```

```
public:
```

```
    Person() : id(0), name("") {}
```

```
    Person(int id, const string& name) : id(id),
```

```
    name(name) {}
```

```
    virtual ~Person() {}
```

```
    virtual void input() {
```

```
        cout << "请输入编号: ";
```

```
        cin >> id;
```

```
        cin.ignore();
```

```
        cout << "请输入姓名: ";
```

```
        getline(cin, name);
```

```
    }
```

```
    virtual void display() const {
```

```
        cout << "编号: " << id << endl;
```

```
        cout << "姓名: " << name << endl;
```

```
    }
```

```
int getId() const { return id; }
```

```
string getName() const { return name; }
```

```
};
```

```
class Student : public Person {
```

```
private:
```

```
    string classNo;
```

```
    double score;
```

```
public:
```

```
    Student() : Person(), classNo(""), score(0.0) {}
```

```
    Student(int id, const string& name, const string&
```

```
    classNo, double score)
```

```
        : Person(id, name), classNo(classNo),
```

```
        score(score) {}
```

```
    virtual void input() {
```

```
        Person::input();
```

```
        cout << "请输入班号: ";
```

```
        getline(cin, classNo);
```

```
        cout << "请输入成绩: ";
```

```
        cin >> score;
```

```
        cin.ignore();
```

```
    }
```

```
    virtual void display() const {
```

```
        cout << "\n=== 学生信息 ===" << endl;
```

```

        Person::display();

        cout << "班号: " << classNo << endl;

        cout << "成绩: " << score << endl;
    }

    string getClassNo() const { return classNo; }

    double getScore() const { return score; }
};

class Teacher : public Person {
private:
    string title;

    string department;

public:
    Teacher() : Person(), title(""), department("") {}

    Teacher(int id, const string& name, const string&
title, const string& department)
        : Person(id, name), title(title),
department(department) {}

    virtual void input() {
        Person::input();

        cout << "请输入职称: ";

        getline(cin, title);

        cout << "请输入部门: ";

        getline(cin, department);
    }
}

```

```

    virtual void display() const {
        cout << "\n=== 教师信息 ===" << endl;

        Person::display();

        cout << "职称: " << title << endl;

        cout << "部门: " << department << endl;
    }

    string getTitle() const { return title; }

    string getDepartment() const { return department; }
};

void showMenu() {
    cout << "\n===== 学生教师管理系统 =====" <<
endl;

    cout << "1. 添加学生" << endl;

    cout << "2. 添加教师" << endl;

    cout << "3. 显示所有人员" << endl;

    cout << "4. 退出" << endl;

    cout << "请输入您的选择: ";
}

int main() {
    vector<Person*> people;

    int choice;

    do {
        showMenu();

        cin >> choice;

        cin.ignore();
    }
}

```

```

                                for (size_t i = 0; i <
switch (choice) {                                people.size(); i++) {
    case 1: {                                    cout << "\n 人员  " << i
        Student* student = new Student();      + 1 << ":" << endl;
        cout << "\n--- 输入学生信息 ---"        people[i]->display();
<< endl;                                        }
        student->input();                        }
        people.push_back(student);              break;
        cout << "学生信息添加成功!" <<        }
endl;                                           case 4: {
        break;                                  for (size_t i = 0; i < people.size();
    }                                           i++) {
    case 2: {                                    delete people[i];
        Teacher* teacher = new Teacher();      }
        cout << "\n--- 输入教师信息 ---"        people.clear();
<< endl;                                       cout << "再见!" << endl;
        teacher->input();                      break;
        people.push_back(teacher);              }
        cout << "教师信息添加成功!" <<        default: {
endl;                                           cout << "\n 无效的选择，请重新输
        break;                                入!" << endl;
    }                                           break;
    case 3: {                                    }
        if (people.empty()) {                  }
            cout << "\n 目前没有人员信        } while (choice != 4);
息!" << endl;
        } else {                                return 0;
            cout << "\n--- 所有人员信息
---" << endl;

```

```
请输入您的选择：1

--- 输入学生信息 ---
请输入编号：001
请输入姓名：张三
请输入班号：03
请输入成绩：90
学生信息添加成功！

===== 学生教师管理系统 =====
1. 添加学生
2. 添加教师
3. 显示所有人员
4. 退出
请输入您的选择：2

--- 输入教师信息 ---
请输入编号：103
请输入姓名：李四
请输入职称：教务主任
请输入部门：教务处
教师信息添加成功！

===== 学生教师管理系统 =====
1. 添加学生
2. 添加教师
3. 显示所有人员
4. 退出
请输入您的选择：3

--- 所有人员信息 ---

人员 1:

=== 学生信息 ===
编号：1
姓名：张三
班号：03
成绩：90

人员 2:

=== 教师信息 ===
编号：103
姓名：李四
职称：教务主任
部门：教务处
}
```

```

#include <iostream>

#include <string>

using namespace std;

class Teacher {
protected:
    string name;

    int age;

    string sex;

    string address;

    string phone;

    string title;
public:
    Teacher(string n, int a, string s, string addr, string p,
string t);

    void display();
};

Teacher::Teacher(string n, int a, string s, string addr,
string p, string t) {

    name = n;

    age = a;

    sex = s;

    address = addr;

    phone = p;

    title = t;
}

void Teacher::display() {

    cout << "姓名: " << name << endl;

    cout << "年龄: " << age << endl;

    cout << "性别: " << sex << endl;

    cout << "职称: " << title << endl;

    cout << "地址: " << address << endl;

    cout << "电话: " << phone << endl;
}

class Cadre {
protected:
    string name;

    int age;

    string sex;

    string address;

    string phone;

    string post;
public:
    Cadre(string n, int a, string s, string addr, string p,
string pos);

    void display();
};

Cadre::Cadre(string n, int a, string s, string addr, string p,
string pos) {

    name = n;

    age = a;

    sex = s;

    address = addr;

    phone = p;

    post = pos;
}

```

```

    }

void Cadre::display() {

    cout << "姓名: " << name << endl;

    cout << "年龄: " << age << endl;

    cout << "性别: " << sex << endl;

    cout << "职务: " << post << endl;

    cout << "地址: " << address << endl;

    cout << "电话: " << phone << endl;

}

class Teacher_Cadre : public Teacher, public Cadre {

private:

    int wages;

public:

    Teacher_Cadre(string n, int a, string s, string addr,

string p, string t, string pos, int w);

    void show();

};

Teacher_Cadre::Teacher_Cadre(string n, int a, string s,

string addr, string p, string t, string pos, int w)

: Teacher(n, a, s, addr, p, t), Cadre(n, a, s, addr, p,

pos) {

    wages = w;

}

void Teacher_Cadre::show() {

    Teacher::display();

    cout << "职务: " << Cadre::post << endl;

    cout << "工资: " << wages << endl;

}

int main() {

    Teacher_Cadre tc("张三", 36, "男", "西安市长安区

", "18712305123", "教师", "教务处主任", 10000);

    cout << "教师兼干部信息: " << endl;

    tc.show();

    return 0;
}

```

```

教师兼干部信息：
姓名：张三
年龄：36
性别：男
职称：教师
地址：西安市长安区
电话：18712305123
职务：教务处主任
工资：10000

```

```
#include <iostream>
```

```
#include <cmath>
```

```
using namespace std;
```

```
class Shape {
```

```
public:
```

```
    virtual double area() = 0;
```

```
    virtual ~Shape() {}
```

```
};
```

```
class Circle : public Shape {
```

```
private:
```

```
    double radius;
```

```
public:
```

```
    Circle(double r) : radius(r) {}
```

```
    double area() {
```

```
        return 3.14159 * radius * radius;
```

```
    }
```

```
};
```

```
class Square : public Shape {
```

```
private:
```

```
    double side;
```

```
public:
```

```
    Square(double s) : side(s) {}
```

```
    double area() {
```

```
        return side * side;
```

```
    }
```

```
};
```

```
class Rectangle : public Shape {
```

```
private:
```

```
    double length;
```

```
    double width;
```

```
public:
```

```
    Rectangle(double l, double w) : length(l), width(w)
```

```
    {}
```

```
    double area() {
```

```
        return length * width;
```

```
    }
```

```
};
```

```
class Trapezoid : public Shape {
```

```
private:
```

```
    double top;
```

```
    double bottom;
```

```
    double height;
```

```
public:
```

```
    Trapezoid(double t, double b, double h) : top(t),
```

```
    bottom(b), height(h) {}
```

```
    double area() {
```

```
        return (top + bottom) * height / 2;
```

```
    }
```

```
};
```

```
class Triangle : public Shape {
```

```
private:
```

```
    double base;
```

```
    double height;
```

```
public:
```

```

Triangle(double b, double h) : base(b), height(h) {}

double area() {
    return base * height / 2;
}

};

```

```

}

```

```

图形1的面积： 50.2654
图形2的面积： 49
图形3的面积： 45
图形4的面积： 22
图形5的面积： 20
总面积： 186.265

```

```

int main() {
    Shape* shapes[5];

    shapes[0] = new Circle(4.0);
    shapes[1] = new Square(7.0);
    shapes[2] = new Rectangle(5.0, 9.0);
    shapes[3] = new Trapezoid(5.0, 6.0, 4.0);
    shapes[4] = new Triangle(8.0, 5.0);

    double totalArea = 0;

    for (int i = 0; i < 5; i++) {
        double area = shapes[i]->area();

        cout << "图形" << i+1 << "的面积: " << area
<< endl;

        totalArea += area;
    }

    cout << "总面积: " << totalArea << endl;

    for (int i = 0; i < 5; i++) {
        delete shapes[i];
    }

    return 0;
}

```